Comprehensive Mechanical Integrity & Fitness-For-Service Assessment Report

Crude Distillation Unit (CDU-1) — Heat Exchanger E-204

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EXECUTIVE SUMMARY

A formal 4-part fitness-for-service evaluation of Heat Exchanger E-204 per API 579-1. Evaluates ultrasonic thickness readings, corrosion rates, remaining safe operating life, and scheduled maintenance requirements.

1. Executive Summary & Facility Context

This comprehensive technical assessment report establishes the mechanical integrity and remaining safe operating lifespan for Heat Exchanger E-204 located within Crude Distillation Unit CDU-1.

All calculations adhere strictly to API 579-1 / ASME FFS-1 Fitness-For-Service standards, ASME Section VIII Division 1 Boiler and Pressure Vessel Code, and plant reliability mandates.

- Assessment Standard: API 579-1 Part 4 (General Metal Loss) Level 1 Assessment
- Operating Pressure: 18.2 bar design / 14.5 bar normal operating threshold
- Operating Temperature: 185 deg C inlet / 130 deg C outlet

2. Ultrasonic Thickness Measurement (UTM) Empirical Data

Ultrasonic thickness inspection was completed on 18 September 2026 across 24 critical Thickness Measurement Locations (TMLs). The nominal specification is SA-516 Gr. 70 carbon steel with nominal wall thickness of 10.0 mm.

Table: Summary of Critical TML Inspection Points

TML Tag	Location / Component	Nominal (mm)	Measured (mm)	Loss (mm)	Corrosion Rate (mm/yr)
TML-01	Shell Course 1 Top	10.0	9.6	0.4	0.08
TML-07	Shell Course 2 Bottom	10.0	9.1	0.9	0.12
TML-14	Channel Side Inlet	10.0	8.2	1.8	0.15
TML-19	Channel Side Return Nozzle	10.0	8.4	1.6	0.14
TML-24	Floating Head Cover	12.0	11.2	0.8	0.09

3. Mathematical Modeling & Code Computations

The minimum required retirement wall thickness (t_min) is determined per ASME Section VIII Division 1 UG-27 equation for cylindrical shells under internal design pressure:

1. Step 1: Compute circumferential hoop stress and minimum thickness requirement (7.0 mm).
2. Step 2: Establish remaining corrosion allowance across governing location TML-14 (1.2 mm).
3. Step 3: Compute remaining operating lifespan based on maximum observed corrosion rate (8.0 years).

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# API 579-1 Level 1 Calculation for Cylindrical Shell
P = 18.2 # Design Pressure (bar)
R = 600.0 # Inside Radius (mm)
S = 138.0 # Allowable Stress for SA-516 Gr.70 (MPa)
E = 1.0 # Joint Efficiency (Full Radiography)

# t_min = (P * R) / (S * E - 0.6 * P)
t_min = (1.82 * 600.0) / (138.0 * 1.0 - 0.6 * 1.82)
# Result: t_min = 7.02 mm (Design Retirement Limit: 7.0 mm)
remaining_ca = 8.2 - 7.0 # Remaining Corrosion Allowance = 1.2 mm
remaining_life = remaining_ca / 0.15 # Remaining Life = 8.0 Years
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4. Engineering Recommendations & Action Plan

Heat Exchanger E-204 is approved for continued safe operation under current design conditions. However, the localized thinning at TML-14 requires increased inspection frequency.

An ultrasonic reinspection is scheduled for Q3 2027 to verify that corrosion rates do not accelerate.

- Next Inspection Date: September 2027 (12-month interval)
- Action Item: Install online ultrasonic corrosion monitoring sensors at TML-14 and TML-19
- Corrosion Inhibitor: Maintain neutralizer dosing at CDU overhead accumulator to minimize acid condensation